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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING(AI&ML)

AGENTIC DEEP RESEARCH SYSTEM

Under the guidance of

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PROBLEM STATEMENT

- Traditional search requires users to manually search, open, compare, and summarize multiple sources.
- General-purpose AI systems may provide answers without sufficiently transparent source verification.
- Information from multiple sources can be difficult to organize into a structured research report.
- AI-generated answers may contain unsupported or inaccurate claims.
- There is a need for an autonomous research system that can plan, search, retrieve, analyze, verify, and cite information.
- The proposed system aims to reduce manual research effort while improving source traceability and reliability.

ABSTRACT

- Agentic Deep Researcher is an AI-based multi-step research system.
- It accepts a broad research question from the user.
- A Planner Agent decomposes the question into smaller research questions.
- Searcher and Fetcher Agents collect relevant information from web sources.
- A Research Agent analyzes the collected information and generates a structured report.
- A Critic Agent is planned to review the report for missing information and unsupported claims.
- The system will ultimately support iterative research, source-based citations, and confidence assessment.
- The current prototype demonstrates the basic research pipeline using a local LLM and web search.

LITERATURE SURVEY

- **Yao et al. (2023) – ReAct: Synergizing Reasoning and Acting in Language Models**
 - Combined reasoning and action in language models
 - Allowed agents to interact with external tools and environments
 - Improved performance on knowledge-intensive tasks
 - Demonstrated iterative reason → act → observe behaviour
 - **Limitation:** Can still produce errors during reasoning and tool selection

LITERATURE SURVEY

- **Schick et al. (2023) – Toolformer: Language Models Can Teach Themselves to Use Tools**
 - Focused on enabling language models to use external tools
 - Used tools such as calculators, search engines, and APIs
 - Improved the ability of LLMs to obtain information beyond their internal knowledge
 - Demonstrated the importance of tool-augmented AI systems
 - **Limitation:** Tool selection and generated tool calls may still be unreliable

LITERATURE SURVEY

➤ Shinn et al. (2023) – Reflexion: Language Agents with Verbal Reinforcement Learning

- Introduced self-reflection for language agents
- Agents evaluate previous actions and learn from their mistakes
- Used verbal feedback to improve subsequent attempts
- Helped agents perform better on tasks requiring multiple attempts
- **Limitation:** Depends on the quality of self-evaluation and may propagate incorrect feedback

LITERATURE SURVEY

➤ Wu et al. (2023) – AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation

- Proposed a framework for multi-agent LLM applications
- Multiple agents can communicate and collaborate on tasks
- Supports specialized roles and automated task execution
- Reduces the need for a single agent to perform every task
- **Limitation:** Multi-agent communication can increase complexity and computational cost

EXISTING SYSTEM

- Traditional search engines return a list of webpages.
- Users must manually:
 - Select relevant sources
 - Open webpages
 - Read information
 - Compare sources
 - Extract important points
 - Prepare the final report
- General AI assistants can summarize information quickly. However, generated answers can contain hallucinated or unsupported information.
- Source verification and research coverage may depend heavily on the user.
- Existing systems generally do not provide the complete autonomous research cycle.

PROPOSED SYSTEM

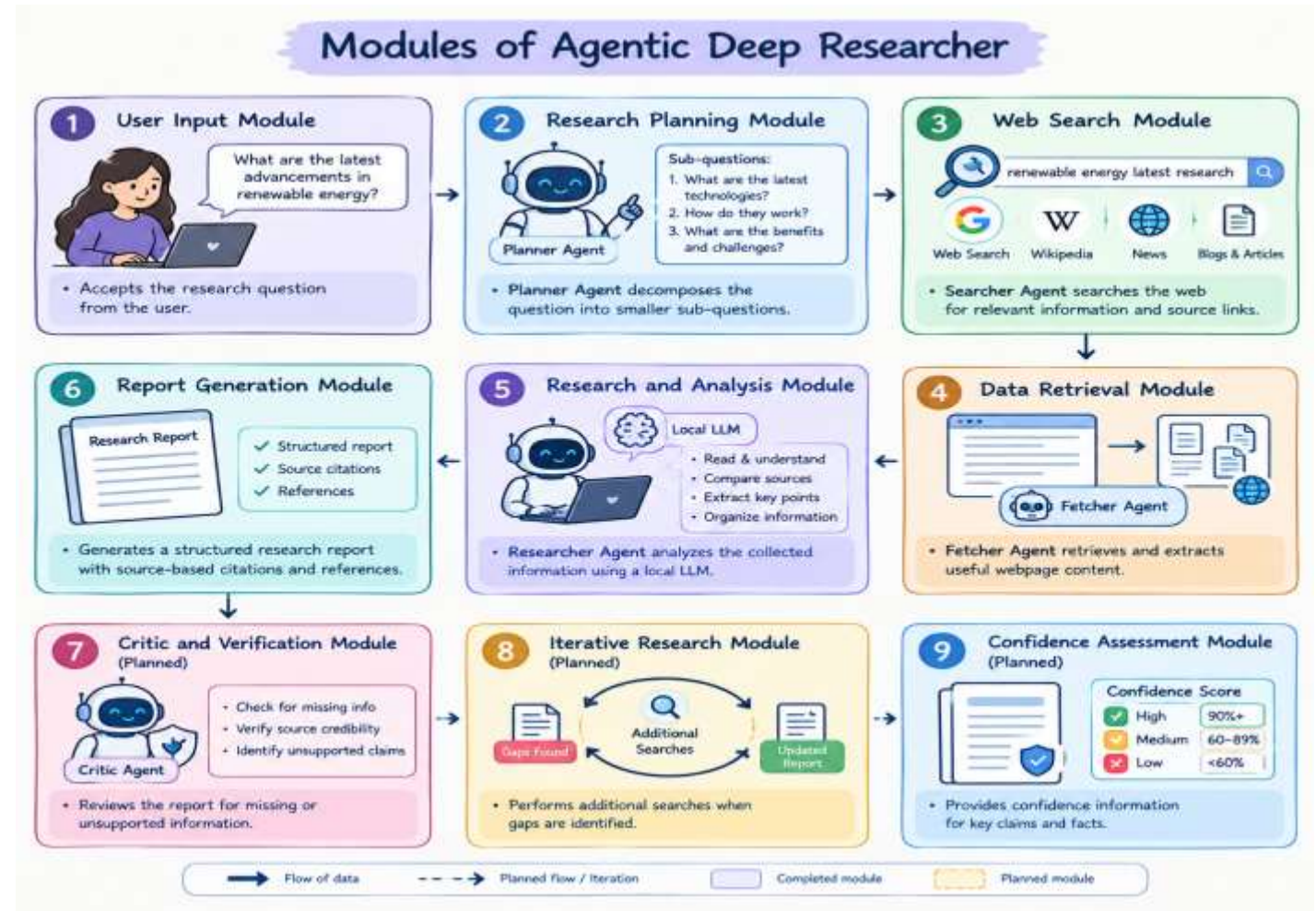
- Develop an Agentic Deep Researcher using multiple specialized agents.
- Planner Agent breaks the main question into sub-questions.
- Searcher Agent searches the web for each sub-question.
- Fetcher Agent retrieves webpage content.
- Research Agent analyzes the collected information.
- Critic Agent checks the generated report.
- If gaps are detected, the system will perform additional searches.
- References will be generated from the actual retrieved source metadata.
- Final system will produce a structured, source-grounded research report.

SYSTEM ARCHITECTURE



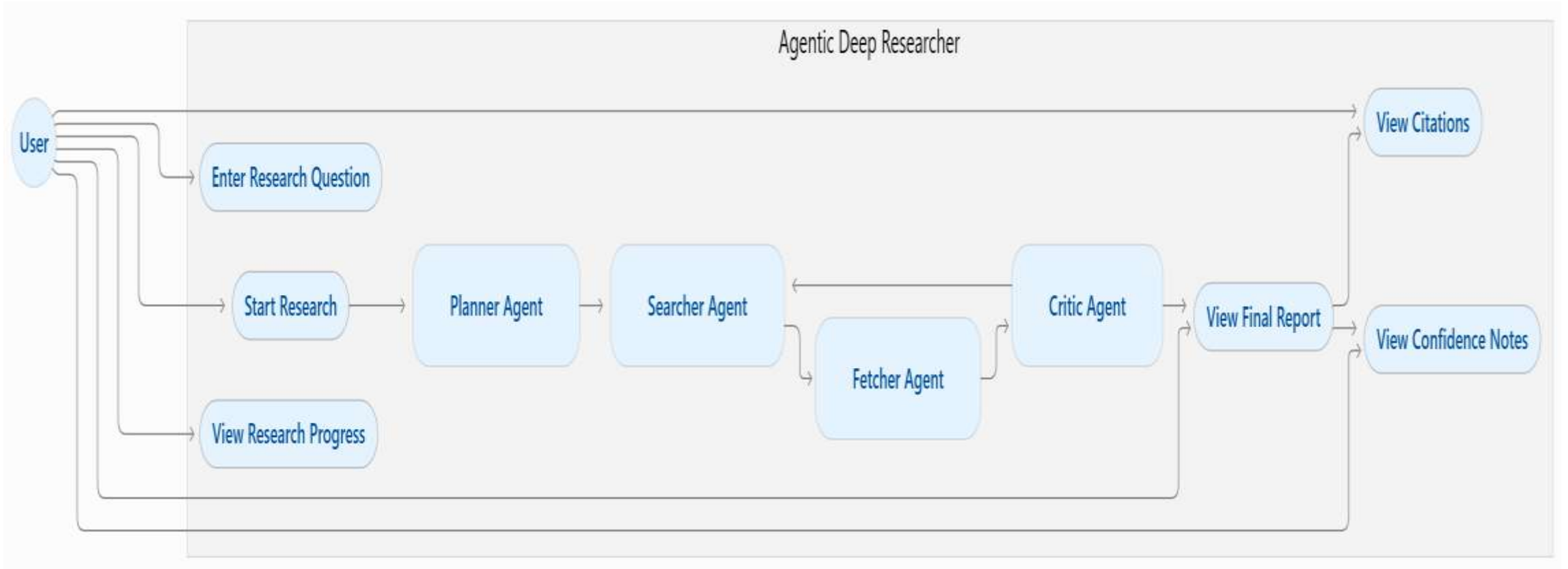
MODULES

- User Input Module
- Research Planning Module
- Web Search Module
- Data Retrieval Module
- Research and Analysis Module
- Report Generation Module
- Critic / Verification Module
- Final Report with Citations and References



UML DIAGRAMS

- Use Case Diagram :



UML DIAGRAMS

- Activity Diagram :

